

Controlled Experiment Submission Report

Submitted Files

Experiment result matrix file: reports/experiment_result_matrix.csv
Plot file: reports/metric_over_time_plot.svg
Memo file: reports/submission_memo.md
Combined PDF report: reports/controlled_experiment_submission_report.pdf

Controlled Experiment Description

The controlled experiment compares four paired runs: treatment prediction without weather, treatment prediction with weather, burnout-index regression without weather, and burnout-index regression with weather. The same random state, split policy, preprocessing pipeline, target, and candidate model order are held fixed within each paired comparison. The only intended treatment difference is whether weather-derived predictors are included.

Error Taxonomy

The error taxonomy separates data/preprocessing failures, model-fit failures, metric/evaluation failures, and report-generation failures. The latest run had no model failures; sklearn warnings that did not stop execution were not counted as failures.

Latest Run Status

88 complete models; 0 failures.

Best Model By Controlled Run rows 1-4 of 4

run_id	target	task_type	model	primary_metric	accuracy	f1_weighted	rmse	mae	r2
burnout_index_no	burnout_index	regression	HistGradientBoos	0.793805271643	nan	nan	0.883133307052	0.708394136876	0.79380527164393
burnout_index_w	burnout_index	regression	HistGradientBoos	0.795840162339	nan	nan	0.878764777546	0.705409091720	0.79584016233940
treatment_non_w	sought_treatmen	classification	HistGradientBoos	0.796165854401	0.795918367346	0.796165854401	nan	nan	nan
treatment_weath	sought_treatmen	classification	Logistic_C0.1	0.796336256021	0.795918367346	0.796336256021	nan	nan	nan

Experiment-Result Matrix rows 1-26 of 88

run_id	model_index	model	status	primary_metric	accuracy	f1_weighted	rmse	mae	r2
burnout_index_n	1	DummyMean	complete	-5.77402842250	nan	nan	1.944912301508	1.592504947121	-5.7740284225005
burnout_index_n	2	LinearRegression	complete	0.781639133707	nan	nan	0.908813751299	0.729993760348	0.78163913370769
burnout_index_n	3	Ridge_alpha1	complete	0.781639596714	nan	nan	0.908812787785	0.729995190680	0.78163959671482
burnout_index_n	4	Ridge_alpha10	complete	0.781642886629	nan	nan	0.908805941470	0.730009742934	0.78164288662977
burnout_index_n	5	Lasso_alpha0001	complete	0.781666440837	nan	nan	0.908756923645	0.729943918295	0.78166644083788
burnout_index_n	6	Lasso_alpha001	complete	0.781760369929	nan	nan	0.908561424851	0.729772187142	0.78176036992951
burnout_index_n	7	ElasticNet_alpha	complete	0.781713352806	nan	nan	0.908659288938	0.729934260472	0.78171335280604
burnout_index_n	8	ElasticNet_alpha	complete	0.777653803776	nan	nan	0.917069685117	0.738369314346	0.77765380377673
burnout_index_n	9	SGDReg_alpha00	complete	0.780889016035	nan	nan	0.910373400718	0.731263095906	0.78088901603568
burnout_index_n	10	SGDReg_alpha00	complete	0.780845258612	nan	nan	0.910464298958	0.731494247811	0.78084525861255
burnout_index_n	11	LinearSVR_C0.5	complete	0.781554311993	nan	nan	0.908990247336	0.729316678764	0.78155431199371
burnout_index_n	12	LinearSVR_C1	complete	0.781553462906	nan	nan	0.908992013934	0.729362610683	0.78155346290661
burnout_index_n	13	KNNReg_k5	complete	0.555054092256	nan	nan	1.297302018643	1.030498293858	0.55505409225622
burnout_index_n	14	KNNReg_k15	complete	0.581923123812	nan	nan	1.257521905882	1.008545832742	0.58192312381212
burnout_index_n	15	DecisionTreeReg	complete	0.725538178484	nan	nan	1.018892712679	0.814534952899	0.72553817848419
burnout_index_n	16	RandomForestRe	complete	0.759995000490	nan	nan	0.952790963882	0.762230216805	0.75999500049006
burnout_index_n	17	RandomForestRe	complete	0.785586066403	nan	nan	0.900562767361	0.720123896947	0.78558606640300
burnout_index_n	18	ExtraTreesReg_d	complete	0.755236981536	nan	nan	0.962188995575	0.771971703636	0.75523698153605
burnout_index_n	19	GradientBoosting	complete	0.758465155862	nan	nan	0.955822789705	0.771897068505	0.75846515586229
burnout_index_n	20	HistGradientBoos	complete	0.793805271643	nan	nan	0.883133307052	0.708394136876	0.79380527164393
burnout_index_n	21	NeuralNetReg_M	complete	0.790363429903	nan	nan	0.890473517588	0.713240299828	0.79036342990375
burnout_index_n	22	NeuralNetReg_M	complete	0.787753427151	nan	nan	0.895999626555	0.717904760774	0.78775342715127
burnout_index_w	1	DummyMean	complete	-5.77402842250	nan	nan	1.944912301508	1.592504947121	-5.7740284225005
burnout_index_w	2	LinearRegression	complete	0.781779650093	nan	nan	0.908521290985	0.729806520653	0.78177965009370
burnout_index_w	3	Ridge_alpha1	complete	0.781780129181	nan	nan	0.908520293687	0.729808028448	0.78178012918117
burnout_index_w	4	Ridge_alpha10	complete	0.781783562848	nan	nan	0.908513145922	0.729823910563	0.78178356284846

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run_id	model_index	model	status	primary_metric	accuracy	f1_weighted	rmse	mae	r2
burnout_index_w	5	Lasso_alpha0001	complete	0.781807938238	nan	nan	0.908462402768	0.729755584180	0.78180793823836
burnout_index_w	6	Lasso_alpha001	complete	0.781907437642	nan	nan	0.908255241760	0.729589324763	0.78190743764215
burnout_index_w	7	ElasticNet_alpha	complete	0.781858620245	nan	nan	0.908356887077	0.729749886686	0.78185862024578
burnout_index_w	8	ElasticNet_alpha	complete	0.777793771626	nan	nan	0.916780990096	0.738144772954	0.77779377162669
burnout_index_w	9	SGDReg_alpha00	complete	0.781021942626	nan	nan	0.910097213760	0.731162917470	0.78102194262695
burnout_index_w	10	SGDReg_alpha00	complete	0.780979463008	nan	nan	0.910185484494	0.731400392428	0.78097946300842
burnout_index_w	11	LinearSVR_C0.5	complete	0.781695943601	nan	nan	0.908695522763	0.729154915108	0.78169594360140
burnout_index_w	12	LinearSVR_C1	complete	0.781734183942	nan	nan	0.908615931138	0.729049787811	0.78173418394294
burnout_index_w	13	KNNReg_k5	complete	0.518727919741	nan	nan	1.349220152682	1.074925824892	0.51872791974167
burnout_index_w	14	KNNReg_k15	complete	0.554931315627	nan	nan	1.297480992513	1.045439384695	0.55493131562789
burnout_index_w	15	DecisionTreeReg	complete	0.725587956303	nan	nan	1.018800312657	0.814468775794	0.72558795630340
burnout_index_w	16	RandomForestRe	complete	0.760002102661	nan	nan	0.952776866395	0.762339346174	0.76000210266142
burnout_index_w	17	RandomForestRe	complete	0.786615242909	nan	nan	0.898398838696	0.718732278988	0.78661524290924
burnout_index_w	18	ExtraTreesReg_d	complete	0.754976053800	nan	nan	0.962701726057	0.772539083416	0.75497605380048
burnout_index_w	19	GradientBoosting	complete	0.758454083309	nan	nan	0.955844698089	0.771928813469	0.75845408330996
burnout_index_w	20	HistGradientBoos	complete	0.795840162339	nan	nan	0.878764777546	0.705409091720	0.79584016233940
burnout_index_w	21	NeuralNetReg_M	complete	0.791312347496	nan	nan	0.888455872571	0.713014506465	0.79131234749657
burnout_index_w	22	NeuralNetReg_M	complete	0.789282323032	nan	nan	0.892766674404	0.716065567582	0.78928232303229
treatment_non_w	1	DummyMostFreq	complete	0.391514500537	0.551020408163	0.391514500537	nan	nan	nan
treatment_non_w	2	Logistic_C0.1	complete	0.789253524227	0.789115646258	0.789253524227	nan	nan	nan
treatment_non_w	3	Logistic_C1	complete	0.775341991805	0.775510204081	0.775341991805	nan	nan	nan
treatment_non_w	4	Logistic_C10	complete	0.789469330709	0.789115646258	0.789469330709	nan	nan	nan
treatment_non_w	5	Ridge_alpha1	complete	0.782312925170	0.782312925170	0.782312925170	nan	nan	nan
treatment_non_w	6	Ridge_alpha10	complete	0.789253524227	0.789115646258	0.789253524227	nan	nan	nan
treatment_non_w	7	SGDLogistic_alph	complete	0.689695110834	0.693877551020	0.689695110834	nan	nan	nan
treatment_non_w	8	SGDLogistic_alph	complete	0.788957628665	0.789115646258	0.788957628665	nan	nan	nan

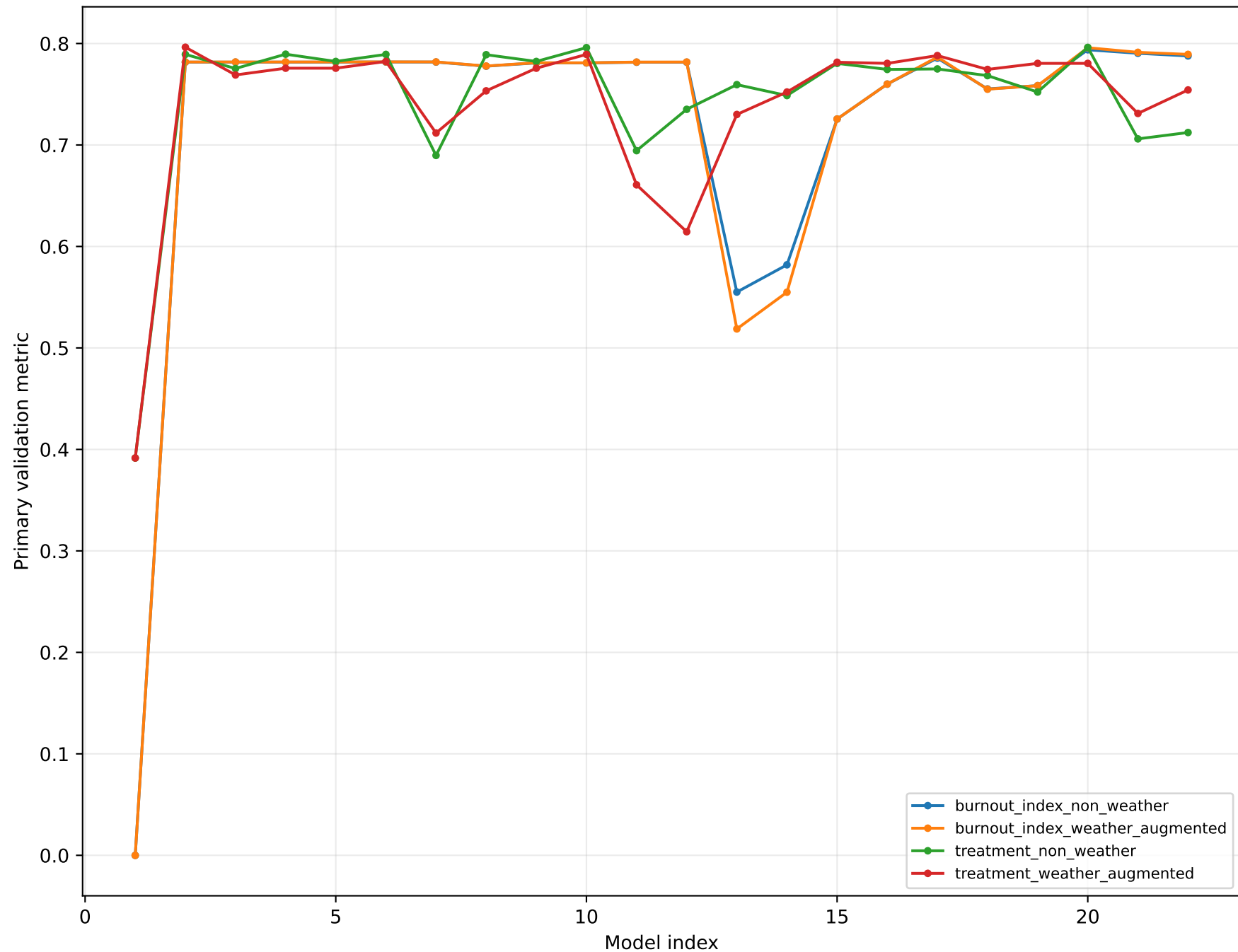
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run_id	model_index	model	status	primary_metric	accuracy	f1_weighted	rmse	mae	r2
treatment_non_w	9	LinearSVC_C0.5	complete	0.782312925170	0.782312925170	0.782312925170	nan	nan	nan
treatment_non_w	10	LinearSVC_C1	complete	0.795918367346	0.795918367346	0.795918367346	nan	nan	nan
treatment_non_w	11	KNN_k5	complete	0.694390963932	0.693877551020	0.694390963932	nan	nan	nan
treatment_non_w	12	KNN_k15	complete	0.735138835408	0.734693877551	0.735138835408	nan	nan	nan
treatment_non_w	13	DecisionTree_dep	complete	0.759459624671	0.761904761904	0.759459624671	nan	nan	nan
treatment_non_w	14	DecisionTree_dep	complete	0.748721459233	0.748299319727	0.748721459233	nan	nan	nan
treatment_non_w	15	RandomForest_d	complete	0.780412810285	0.782312925170	0.780412810285	nan	nan	nan
treatment_non_w	16	RandomForest_fu	complete	0.774451739298	0.775510204081	0.774451739298	nan	nan	nan
treatment_non_w	17	ExtraTrees_depth	complete	0.774940673943	0.775510204081	0.774940673943	nan	nan	nan
treatment_non_w	18	ExtraTrees_full	complete	0.768338630011	0.768707482993	0.768338630011	nan	nan	nan
treatment_non_w	19	GradientBoosting	complete	0.752184365930	0.755102040816	0.752184365930	nan	nan	nan
treatment_non_w	20	HistGradientBoos	complete	0.796165854401	0.795918367346	0.796165854401	nan	nan	nan
treatment_non_w	21	NeuralNet_MLP_3	complete	0.706015037593	0.714285714285	0.706015037593	nan	nan	nan
treatment_non_w	22	NeuralNet_MLP_6	complete	0.712279122767	0.721088435374	0.712279122767	nan	nan	nan
treatment_weath	1	DummyMostFreq	complete	0.391514500537	0.551020408163	0.391514500537	nan	nan	nan
treatment_weath	2	Logistic_C0.1	complete	0.796336256021	0.795918367346	0.796336256021	nan	nan	nan
treatment_weath	3	Logistic_C1	complete	0.768987968321	0.768707482993	0.768987968321	nan	nan	nan
treatment_weath	4	Logistic_C10	complete	0.775656977403	0.775510204081	0.775656977403	nan	nan	nan
treatment_weath	5	Ridge_alpha1	complete	0.775656977403	0.775510204081	0.775656977403	nan	nan	nan
treatment_weath	6	Ridge_alpha10	complete	0.782312925170	0.782312925170	0.782312925170	nan	nan	nan
treatment_weath	7	SGDLogistic_alph	complete	0.711791813499	0.714285714285	0.711791813499	nan	nan	nan
treatment_weath	8	SGDLogistic_alph	complete	0.753352769679	0.755102040816	0.753352769679	nan	nan	nan
treatment_weath	9	LinearSVC_C0.5	complete	0.775656977403	0.775510204081	0.775656977403	nan	nan	nan
treatment_weath	10	LinearSVC_C1	complete	0.789253524227	0.789115646258	0.789253524227	nan	nan	nan
treatment_weath	11	KNN_k5	complete	0.660717129661	0.659863945578	0.660717129661	nan	nan	nan
treatment_weath	12	KNN_k15	complete	0.614509013670	0.619047619047	0.614509013670	nan	nan	nan

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run_id	model_index	model	status	primary_metric	accuracy	f1_weighted	rmse	mae	r2
treatment_weath	13	DecisionTree_dep	complete	0.730056594066	0.734693877551	0.730056594066	nan	nan	nan
treatment_weath	14	DecisionTree_dep	complete	0.752184365930	0.755102040816	0.752184365930	nan	nan	nan
treatment_weath	15	RandomForest_d	complete	0.781534300467	0.782312925170	0.781534300467	nan	nan	nan
treatment_weath	16	RandomForest_fu	complete	0.780412810285	0.782312925170	0.780412810285	nan	nan	nan
treatment_weath	17	ExtraTrees_depth	complete	0.788121330855	0.789115646258	0.788121330855	nan	nan	nan
treatment_weath	18	ExtraTrees_full	complete	0.774451739298	0.775510204081	0.774451739298	nan	nan	nan
treatment_weath	19	GradientBoosting	complete	0.780412810285	0.782312925170	0.780412810285	nan	nan	nan
treatment_weath	20	HistGradientBoos	complete	0.780412810285	0.782312925170	0.780412810285	nan	nan	nan
treatment_weath	21	NeuralNet_MLP_3	complete	0.731069096056	0.734693877551	0.731069096056	nan	nan	nan
treatment_weath	22	NeuralNet_MLP_6	complete	0.754226088025	0.755102040816	0.754226088025	nan	nan	nan

Metric-Over-Time Plot



Classification primary metric is weighted F1. Burnout-index regression primary metric is R-squared.

Error Taxonomy and Failure Analysis Memo

Error Taxonomy

No model failures in the latest run.

Failure Analysis Memo

No model failures in the latest run.

Warnings from sklearn that do not stop a run are not counted as failures.